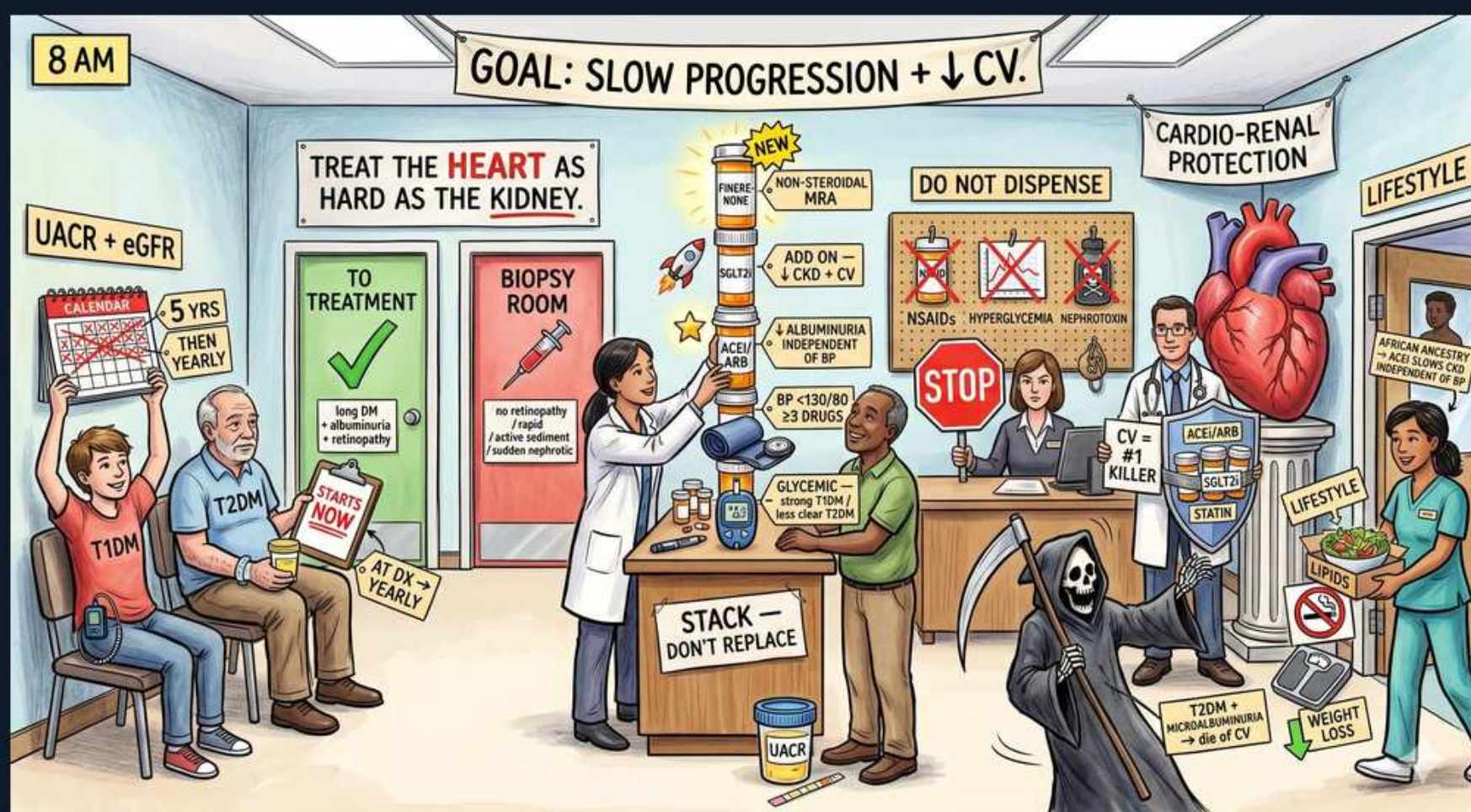


Glomerular (Nephrotic) — Diabetic Nephropathy: Management



1. Goal: slow progression + ↓CV

WHAT YOU SEE

Top banner reading GOAL: SLOW PROGRESSION + ↓CV stretched across the clinic ceiling — the two goals are written together because they are inseparable: protect the kidney AND the heart at once.

WHAT IT ENCODES

The two management goals are inseparable: SLOW renal progression (delay ESKD) AND reduce CARDIOVASCULAR events, because CV disease is the leading killer in diabetic kidney disease. The modern 'pillar' approach stacks RAAS blockade + SGLT2 inhibitor + nonsteroidal MRA (finerenone) + GLP-1 receptor agonist, layered on glycemic, BP, and lipid control — 'treat the heart as hard as the kidney.'

BOARD TRAP

Wrong answer: 'ESRD/dialysis is the most common outcome, so focus only on renal protection.' Discriminator — cardiovascular death is more common than reaching dialysis; the correct strategy is cardio-renal (SGLT2i, finerenone, statins, BP), not kidney-only. Neglecting CV risk is the tested error.

2. Screen: UACR + eGFR

WHAT YOU SEE

Left figure holding up a wall calendar with a tag 'UACR + eGFR' — the calendar plus the two-test tag means screening is a scheduled pairing of urine albumin-creatinine ratio and eGFR, both checked together.

WHAT IT ENCODES

Screening uses TWO measures together: urine ALBUMIN-CREATININE RATIO (UACR, spot urine) and estimated GFR. UACR detects albuminuria — the earliest marker — long before a dipstick turns positive or creatinine rises. Both are needed because a patient can have low eGFR with normal UACR or vice versa; the combination stages risk (KDIGO heat map).

BOARD TRAP

Wrong answer: 'a normal urine dipstick rules out early diabetic kidney disease.' Discriminator — the dipstick only detects MACROalbuminuria (>300 mg/day) and misses microalbuminuria (30–300 mg/g). You must order a quantitative UACR; relying on the dipstick misses the treatable early window.

3. T1DM screen 5 yrs then yearly

WHAT YOU SEE

Younger 'T1DM' patient holding a calendar marked '5 YRS THEN YEARLY' — the crossed-off years on the calendar mean T1DM screening starts 5 years after diagnosis, then annually.

WHAT IT ENCODES

Type 1 DM: begin annual screening (UACR + eGFR) 5 YEARS after diagnosis, because the onset date is known and clinically significant nephropathy rarely develops before then. After the 5-year mark, screen every year.

BOARD TRAP

Wrong answer: 'screen T1DM patients immediately at diagnosis like T2DM.' Discriminator — T1DM has a KNOWN onset, so screening starts at 5 years; T2DM has an UNKNOWN (often years-long preclinical) onset, so screening starts at diagnosis. Swapping the two timelines is the classic mix-up.

4. T2DM screen at diagnosis

WHAT YOU SEE

Older 'T2DM' patient holding a 'STARTS NOW' sign with 'A1c / yearly' — the STARTS NOW placard means T2DM screening begins immediately at diagnosis (no grace period), because the disease was silent for years.

WHAT IT ENCODES

Type 2 DM: screen with UACR + eGFR AT THE TIME OF DIAGNOSIS, then yearly. Because type 2 diabetes is often present (and silently damaging the kidney) for years before it is detected, nephropathy may already exist at diagnosis — so there is no 5-year grace period.

BOARD TRAP

Wrong answer: 'wait 5 years after T2DM diagnosis to start screening.' Discriminator — the 5-year delay applies only to T1DM (known onset). In T2DM the true disease onset predates diagnosis, so you screen NOW and yearly thereafter.

5. Biopsy red flags

WHAT YOU SEE

Red BIOPSY ROOM door listing 'no retinopathy / rapid / active sediment / sudden nephrotic' — the door only opens (biopsy only indicated) for these atypical red-flag features, not for typical diabetic nephropathy.

WHAT IT ENCODES

Diabetic nephropathy is usually a CLINICAL diagnosis (long-standing diabetes + albuminuria + retinopathy, bland sediment) and does NOT require biopsy. Biopsy is reserved for ATYPICAL features: absent retinopathy (especially in T1DM), rapid GFR decline, active urinary sediment (RBC/cellular casts, hematuria), sudden onset of nephrotic-range proteinuria, or short diabetes duration — all suggest a superimposed/alternative glomerular disease.

BOARD TRAP

Wrong answer: 'every diabetic with proteinuria needs a confirmatory renal biopsy.' Discriminator — typical diabetic nephropathy is diagnosed clinically. The trap is BOTH over-biopsying typical cases AND ignoring red flags (no retinopathy, active sediment, rapid decline) that actually mandate biopsy.

6. ACEi/ARB cornerstone

WHAT YOU SEE

Base block of the stacked drug tower labelled 'ACEi/ARB' — it is drawn as the foundation the whole stack rests on, because RAAS blockade is the cornerstone of therapy.

WHAT IT ENCODES

RAAS blockade with an ACE inhibitor OR an ARB is the FOUNDATION (base of the tower): by dilating the efferent arteriole it lowers intraglomerular pressure, reducing albuminuria and slowing progression — indicated for diabetics with hypertension and/or albuminuria. Titrate to the maximally tolerated dose; a ≤30% creatinine rise and mild potassium elevation are acceptable and expected.

BOARD TRAP

Wrong answer: 'combine an ACE inhibitor WITH an ARB for additive proteinuria reduction.' Discriminator — dual RAAS blockade (ACEi + ARB, or with direct renin inhibitor) increases hyperkalemia, AKI, and hypotension WITHOUT outcome benefit and is contraindicated. Use one or the other, not both.

7. SGLT2 inhibitor

WHAT YOU SEE

'SGLT2' segment stacked in the drug tower with the tag 'albuminuria-independent' — its place built into the tower shows it is a core additive pillar whose benefit is glucose-independent.

WHAT IT ENCODES

SGLT2 inhibitors (empagliflozin, dapagliflozin, canagliflozin) are a core pillar: they slow CKD progression, reduce heart-failure hospitalizations and CV death, and lower albuminuria — benefits that are largely GLUCOSE-INDEPENDENT (they help even at lower eGFR where glucose-lowering effect is small). Expect a small, reversible initial eGFR dip; watch for genital mycotic infections and rare euglycemic DKA.

BOARD TRAP

Wrong answer: 'SGLT2i should be stopped once eGFR is low because they no longer lower glucose.' Discriminator — renal/CV protection is glucose-INDEPENDENT and persists at reduced eGFR (continue down to thresholds where evidence supports them). Stopping for the early eGFR dip or for modest CKD is the error.

8. Non-steroidal MRA (finerenone)

WHAT YOU SEE

Gold-star-topped 'FINERENONE / NON-STEROIDAL MRA' segment with a rocket and a 'NEW — add-on ↓CKD + CV' tag crowning the tower — the rocket and NEW flag mark it as the newest pillar bolted on top of a maxed ACEi/ARB.

WHAT IT ENCODES

Finerenone, a NONSTEROIDAL selective mineralocorticoid-receptor antagonist, is added on TOP of a maxed ACEi/ARB (and ideally SGLT2i) for type 2 diabetics with persistent albuminuria; it further reduces CKD progression and CV events (FIDELIO/FIGARO trials). It carries less gynecomastia than steroidal MRAs but still requires potassium monitoring.

BOARD TRAP

Wrong answer: 'finerenone is interchangeable with spironolactone / causes the same gynecomastia.' Discriminator — finerenone is a NONSTEROIDAL MRA with a distinct structure, proven cardio-renal outcome data in diabetic CKD, and far less anti-androgen effect (gynecomastia) than steroidal spironolactone. Still, both raise potassium — monitor it.

9. Stack don't replace

WHAT YOU SEE

Placard at the foot of the drug tower reading 'STACK — DON'T REPLACE' — the tower is literally built up in layers to show the pillars are additive, not interchangeable.

WHAT IT ENCODES

The therapies are ADDITIVE PILLARS, not alternatives: layer RAAS blockade + SGLT2 inhibitor + finerenone + (often) a GLP-1 receptor agonist, each with independent renal/CV benefit. Build the stack up rather than swapping one agent for another — combined use yields the greatest reduction in progression and events.

BOARD TRAP

Wrong answer: 'pick the single best agent and switch to it (either/or).' Discriminator — the pillars work through different mechanisms and their benefits stack; the modern standard is combination therapy, not choosing one drug. Treating them as mutually exclusive is the conceptual trap.

10. Glycemic A1c <7

WHAT YOU SEE

Tower segment labelled 'GLYCEMIC — strong T2DM / less clear T1DM, A1c <7%' — its slot in the stack flags glycemic control as one of the layered pillars with an individualized A1c target.

WHAT IT ENCODES

Glycemic control with an individualized target (commonly A1c <7%) slows microvascular complications including nephropathy. Prefer agents with proven renal/CV benefit — SGLT2 inhibitors and GLP-1 receptor agonists (e.g., semaglutide, liraglutide) — as the 'renal-friendly' glucose-lowering drugs.

BOARD TRAP

Wrong answer: 'drive A1c as low as possible (e.g., <6.5%) in everyone with CKD.' Discriminator — targets are INDIVIDUALIZED; overly tight control causes dangerous HYPOGLYCEMIA, especially in advanced CKD (reduced insulin clearance, impaired gluconeogenesis). Looser targets are appropriate in frail/advanced-CKD patients.

11. BP target ~130/80

WHAT YOU SEE

Reception desk sign 'BP <130/80, ≥3 drugs' — the front-desk target board sets the blood-pressure goal, with ACEi/ARB preferred to reach it.

WHAT IT ENCODES

Blood-pressure control to roughly <130/80 mmHg slows progression of diabetic kidney disease and reduces CV risk; an ACEi or ARB is the preferred first-line antihypertensive precisely because of its additional albuminuria-lowering, renoprotective effect.

BOARD TRAP

Wrong answer: 'a calcium-channel blocker or thiazide is the preferred first-line BP drug in diabetic nephropathy.' Discriminator — in a diabetic with albuminuria the first-line agent is an ACEi/ARB (renoprotection beyond BP). Other agents are add-ons to reach the <130/80 target, not the preferred first choice.

12. Do not dispense NSAIDs

WHAT YOU SEE

Pharmacy counter with a red 'STOP / DO NOT DISPENSE' sign over crossed-out NSAIDs, hyperglycemia agents, and nephrotoxins — the barred pharmacy shelf means these drugs must be avoided as kidney-harming.

WHAT IT ENCODES

Avoid nephrotoxins: NSAIDs constrict the afferent arteriole (inhibiting protective prostaglandins), reducing renal perfusion and accelerating GFR decline and AKI; they also blunt the effect of RAAS blockers and raise potassium. Also avoid/limit other nephrotoxins (IV contrast when avoidable, aminoglycosides) and agents worsening hyperglycemia.

BOARD TRAP

Wrong answer: 'short courses of NSAIDs are safe in diabetic CKD.' Discriminator — even short NSAID use can precipitate AKI in CKD, especially combined with ACEi/ARB and a diuretic (the 'triple whammy'). Acetaminophen is the safer analgesic; reflexively reaching for NSAIDs is the trap.

13. Grim reaper = CV #1 killer

WHAT YOU SEE

Grim reaper with 'CV #1 KILLER' shield

WHAT IT ENCODES

The grim reaper is cardiovascular disease — the LEADING cause of death in diabetic nephropathy, which is why every therapeutic pillar (SGLT2i, finerenone, statins, BP and glycemic control) is chosen for combined cardio-renal benefit, not kidney protection alone. High-intensity statin therapy is indicated for these high-CV-risk patients.

BOARD TRAP

Wrong answer: 'progression to ESRD is the dominant cause of death in diabetic nephropathy.' Discriminator — most patients die of CARDIOVASCULAR disease before reaching dialysis. The high-yield consequence: don't omit statins and CV-protective agents while focusing on the kidney.

14. Lifestyle / weight loss

WHAT YOU SEE

Right-side scrubbed figure carrying a salad bowl past a 'LIFESTYLE / WEIGHT LOSS' sign with a crossed-out cigarette and a scale — the diet, scale, and no-smoking cues show lifestyle measures as adjuncts to the drug pillars.

WHAT IT ENCODES

Lifestyle measures are adjuncts to the drug pillars: weight loss, a moderate-protein/low-sodium diet, smoking cessation, and exercise. Importantly, a type 2 diabetic with albuminuria is already a CARDIOVASCULAR DISEASE EQUIVALENT, so aggressive global risk-factor modification (lipids, BP, smoking) is mandatory alongside renal therapy.

BOARD TRAP

Wrong answer: 'a very high-protein diet improves albuminuria and renal outcomes.' Discriminator — excessive protein intake increases intraglomerular pressure and proteinuria; guidelines favor moderate protein restriction (~0.8 g/kg/day) in CKD, not high-protein diets. Lifestyle is additive, never a substitute for RAAS/SGLT2i therapy.
